



**JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND
TECHNOLOGY**

**DEPARTMENT OF TELECOMMUNICATION AND
INFORMATION ENGINEERING**

ENE221-0220/2021 – Mugwe Angela Muthoni

ETI 2501: PROJECT 1

**INTELLIGENT RIS-BASED BATTERY-FREE 6G INDOOR
NETWORK**

Supervisor

Mr. Kevin Mwongera

A final year project report submitted to the Department of Telecommunication and Information Engineering in partial fulfillment of the requirements for the award of the Degree of Bachelor of Science in Telecommunication and Information Engineering at Jomo Kenyatta University of Agriculture and Technology.

MAY 2026

DECLARATION

This project is my original work, with proper acknowledgments given where necessary. To the best of my knowledge, it has not been submitted to Jomo Kenyatta University of Agriculture and Technology or any other institution for the award of a degree. This proposal adheres to all the criteria set by the Department of Telecommunication and Information Engineering.

Name: Mugwe Angela Muthoni

Registration Number: ENE221-0220/2021

Signature: _____

Date: _____

SUPERVISOR'S CONFIRMATION

This is to certify that the project titled 'Intelligent RIS-Based Battery-Free 6G Indoor Sensor Network Project Proposal' carried out by the above-named student, has been read and approved for meeting part of the requirements and regulations governing the award of Bachelor of Science in Telecommunication and Information Engineering at JKUAT, Kenya.

Supervisor's Name: Mr. Kevin Mwongera

Signature: _____

Date: _____

DEDICATION

Special dedication of this piece of work goes to my beloved parents, family, lecturers, and friends who have strongly encouraged and supported me throughout my entire journey of learning.

ACKNOWLEDGEMENT

The successful completion of this project proposal has been made possible through the support and encouragement of several individuals and institutions. I give thanks to God for His grace, strength, and provision throughout the entire process. His guidance has been instrumental in enabling me to overcome challenges and remain steadfast in achieving this milestone. I sincerely appreciate the Department of Telecommunication and Information Engineering for providing the academic environment and essential resources that facilitated the development of this work.

I am deeply grateful to my family and friends for their continuous encouragement, patience, and moral support throughout the preparation of this proposal. Their understanding and motivation have helped me remain focused and committed to achieving this milestone.

I also acknowledge my dedication, perseverance, and determination, which have played a vital role in bringing this project proposal to its successful completion. To everyone who contributed directly or indirectly to this work, I extend my heartfelt thanks.

ABSTRACT

This project investigated the use of an intelligent Reconfigurable Intelligent Surface (RIS) to enhance battery-free indoor backscatter communication inspired by future 6G wireless communication systems. Conventional indoor wireless sensor networks rely heavily on battery-powered devices, resulting in limited operational lifetime, high maintenance costs, and environmental concerns associated with battery replacement and disposal. Although ambient backscatter communication offers a sustainable alternative by enabling sensor nodes to harvest energy from surrounding radio-frequency signals and communicate without dedicated power sources, indoor environments introduce significant challenges such as severe path loss, signal blockage, multipath fading, and shadowing, which greatly weaken communication reliability. To address these limitations, this project developed and implemented an RIS-assisted battery-free indoor sensor network capable of intelligently manipulating the wireless propagation environment to improve signal transmission. The implemented system consisted of an ambient RF signal source, a 3×3 RIS prototype constructed using aluminium foil reflecting elements, battery-free backscatter sensor nodes, and a receiver unit. Two RIS elements were integrated with PIN diodes to enable dynamic switching between reflection states, while a Bluetooth-capable microcontroller was used to control RIS configurations and implement intelligent RIS operation through adaptive selection of the reflection pattern that maximized received signal strength. A separate backscatter sensor node was implemented to transmit sensor data by modulating ambient RF reflections without generating an active RF signal. In addition to the hardware implementation, MATLAB simulations were conducted to evaluate a 6G-inspired indoor communication environment operating at 28 GHz, incorporating path loss, wall attenuation, multipath fading, and shadowing effects. System performance was evaluated using received signal strength indicator (RSSI), signal-to-noise ratio (SNR), and link reliability under three scenarios: without RIS, with non-intelligent RIS, and with intelligent RIS control. The results showed that the introduction of RIS improved signal strength, SNR, and communication reliability compared to the baseline system without RIS. Furthermore, intelligent RIS control achieved the highest performance by adaptively selecting the optimal reflection configuration based on measured RSSI values. Both the practical implementation and simulation results confirmed that RIS-assisted communication can significantly enhance battery-free backscatter communication in indoor environments. The project therefore demonstrated the feasibility of integrating RIS, intelligent control, and low-power communication technologies as a sustainable solution for future green and smart 6G-enabled IoT networks, with potential applications in smart buildings, environmental monitoring, and healthcare systems.

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LIST OF ABBREVIATIONS

6G	Sixth Generation
AI	Artificial Intelligence
ANN	Artificial Neural Network
AP	Access Point
BER	Bit Error Rate
CSI	Channel State Information
dB	Decibel
EM	Electromagnetic
GHz	Gigahertz
IoT	Internet of Things
PIN	Positive-Intrinsic-Negative (Diode)
RIS	Reconfigurable Intelligent Surface
RF	Radio Frequency
RSSI	Received Signal Strength Indicator
SNR	Signal-to-Noise Ratio
THz	Terahertz
UHF	Ultra High Frequency
Wi-Fi	Wireless Fidelity
WSN	Wireless Sensor Network

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CHAPTER ONE

1. INTRODUCTION

1.1. Background Information

Smart indoor applications such as building automation, environmental monitoring, healthcare systems, and industrial sensing has been growing rapidly leading to the extensive deployment of wireless sensor networks (WSNs). These networks form a core component of modern Internet of Things (IoT) ecosystems by enabling continuous data collection, real-time monitoring, and intelligent decision-making [1]. Indoor sensor networks rely on battery-powered sensor nodes, which introduce critical limitations including finite operational lifetime, frequent battery replacement, increased maintenance costs, and environmental issues related to battery disposal [2] [3].

As wireless communication technologies evolve toward sixth-generation (6G) networks, there is a strong emphasis on intelligence, adaptability, and energy efficiency [1], [4]. A key concept in the 6G vision is the transformation of the wireless propagation environment from a passive medium into an actively programmable entity. Reconfigurable Intelligent Surfaces (RIS) have emerged as a promising enabling technology to achieve this vision [5], [6]. RIS are composed of low-cost, passive or semi-passive elements that can dynamically control the reflection, phase, and direction of electromagnetic waves, thereby enhancing signal propagation without requiring active radio transmission.

In parallel, battery-free communication techniques based on ambient backscatter have gained increasing attention as a sustainable solution for low-power IoT applications [7], [8]. Ambient backscatter systems enable sensor nodes to harvest energy from existing radio frequency (RF) signals such as Wi-Fi or cellular

transmissions and communicate by modulating and reflecting these signals [2], this approach eliminates the need for batteries, significantly reducing maintenance requirements and improving long-term sustainability.

This project proposes the integration of Reconfigurable Intelligent Surfaces with battery-free ambient backscatter communication to address the limitations of conventional indoor sensor networks [10], [11]. By strategically deploying RIS within an indoor environment, the wireless propagation conditions can be intelligently shaped to enhance signal strength, improve coverage, and mitigate the effects of fading and blockage at the locations of battery-free sensor nodes [4], [6].

To further enhance adaptability, artificial intelligence (AI) techniques are incorporated for dynamic RIS control. AI-based optimization enables autonomous adjustment of RIS reflection patterns in response to environmental changes, improving communication reliability without requiring explicit channel state information [12]. This approach aligns closely with core 6G principles, including intelligent radio environments, autonomous operation, and energy-efficient connectivity [1], [4].

1.2. Problem Statement

Despite their potential advantages, battery-free ambient backscatter communication systems face significant challenges in indoor environments. Indoor wireless channels are highly dynamic and complex due to severe signal attenuation, multipath fading, shadowing, and blockage caused by walls, furniture, and human movement [3], [9]. These factors lead to weak backscattered signals and unreliable communication links, limiting the practical deployment of battery-free sensor networks.

Conventional indoor environments are not designed to support battery-free communication, as the wireless propagation conditions are largely uncontrollable [4], [10]. Without intelligent adaptation, ambient backscatter systems struggle to maintain consistent performance, especially in large or densely populated indoor spaces.

1.3. Problem Justification

The justification of this project lies in its potential to address critical challenges associated with energy sustainability, communication reliability, and adaptability in indoor Internet of Things (IoT) environments. Conventional battery-powered wireless sensor networks (WSNs) face limitations such as finite operational lifetime, frequent battery replacement, and high maintenance costs, which restrict their scalability and long-term deployment, particularly in large or densely populated indoor spaces [2], [3]. By exploring battery-free ambient backscatter communication, this study aims to provide a sustainable alternative that reduces energy consumption and environmental impact, thereby supporting green IoT initiatives.

Integrating Reconfigurable Intelligent Surfaces (RIS) with battery-free communication introduces a transformative approach to indoor wireless systems. RIS technology enables intelligent shaping of the wireless propagation environment, enhancing signal strength, mitigating multipath fading, and extending coverage without increasing power consumption [10], [11]. The use of artificial intelligence (AI) for RIS control further improves adaptability, allowing the system to dynamically respond to changes in the indoor environment, such as human movement, furniture rearrangement, or interference from other wireless devices [12]. This aligns with the principles of next-generation 6G networks, including intelligent, autonomous, and energy-efficient connectivity [4], [9].

1.4. Objectives of the Project

The primary objective of this project is to design and implement an intelligent RIS-assisted battery-free indoor sensor network inspired by 6G communication concepts.

1.4.1 General Objectives

To develop an intelligent, RIS-assisted, battery-free indoor sensor network that reliably collects and transmits data with minimal energy consumption, ensuring high coverage, accurate sensing, and low-latency communication in indoor environments.

1.4.2 Specific Objectives

- i. To design and implement a battery-free indoor sensor node prototype capable of monitoring environmental parameters such as temperature and humidity.
- ii. To develop RIS-assisted communication that enables reliable and low-latency data transmission, targeting an average end-to-end latency of ≤ 50 ms.
- iii. To evaluate network performance by measuring energy harvesting efficiency, data accuracy, coverage, and transmission reliability under indoor conditions.

CHAPTER TWO

2. LITERATURE REVIEW

2.1. 6G Wireless Communication and Smart Radio Environments

Sixth-generation (6G) wireless networks are envisioned to surpass the capabilities of 5G by enabling intelligent connectivity, ultra-low latency, massive machine-type communication, and enhanced energy sustainability [1]. A defining feature of 6G is the concept of smart radio environments, where the wireless propagation medium is no longer passive but can be dynamically programmed and controlled to improve communication performance [4].

This paradigm shift moves wireless system design from traditional network-centric optimization toward environment-aware and AI-driven communication. By shaping electromagnetic wave propagation, smart radio environments aim to address challenges in dense and indoor scenarios, enabling emerging applications such as smart buildings, digital twins, and autonomous systems.

2.2. Reconfigurable Intelligent Surfaces (RIS)

Reconfigurable Intelligent Surfaces have emerged as a key enabling technology for smart radio environments. An RIS is typically composed of a large number of passive reflecting elements whose electromagnetic response can be electronically controlled to alter the phase, amplitude, or polarization of incident signals. Unlike active relays or base stations, RIS do not require radio frequency chains or power-hungry components, making them highly energy-efficient [1], [4], [13].

Several studies have demonstrated that RIS can significantly enhance signal strength, coverage, and spectral efficiency in indoor environments. RIS-assisted communication systems have been shown to outperform conventional relay-based solutions while

consuming significantly less power [6]. Other studies have analyzed RIS deployment strategies for indoor coverage enhancement, highlighting their effectiveness in mitigating signal blockage and shadowing caused by walls and obstacles [13].

Most RIS-related studies focus on enhancing conventional battery-powered wireless communication systems. Experimental implementations are still limited, and the integration of RIS with ultra-low-power or battery-free devices remains an open research area. Reconfigurable Intelligent Surfaces (RIS) are a key enabling technology for smart radio environments. An RIS consists of a large number of passive reflecting elements whose electromagnetic properties such as phase, amplitude, or polarization can be electronically adjusted [5]. Unlike active relays or base stations, RIS operates without radio-frequency chains, resulting in very low power consumption.

Previous research has demonstrated that RIS can significantly enhance signal coverage, spectral efficiency, and reliability, particularly in indoor environments. RIS-assisted communication systems have been shown to outperform conventional relay-based solutions while consuming considerably less energy [2]. Other works investigated optimal RIS placement and deployment strategies, confirming their ability to mitigate signal blockage and shadowing caused by walls and obstacles [13].

Despite these promising results, most prior studies focus on conventional battery-powered communication systems, and experimental RIS implementations remain limited. The indoor RIS-assisted communication scenario shown below is reproduced from Wireless Pi [14].

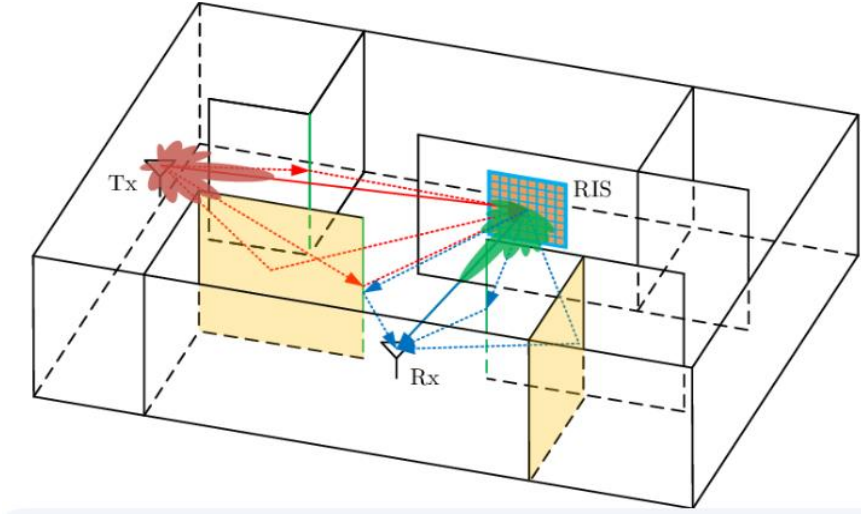


Figure 2.1 Indoor RIS-assisted wireless communication scenario

2.3. Battery-Free Communication and Ambient Backscatter

Battery-free communication has emerged as a sustainable solution for Internet of Things (IoT) networks. Ambient backscatter communication enables ultra-low-power devices to transmit data by modulating and reflecting existing radio-frequency signals from ambient sources such as Wi-Fi access points, cellular base stations, or TV transmitters [2]. This eliminates the need for batteries and significantly reduces maintenance requirements.

Early work done has demonstrated the feasibility of ambient backscatter for short-range indoor communication [7]. Subsequent studies explored improved modulation schemes, receiver architectures, and energy-harvesting techniques [8]. However, the literature consistently reports that ambient backscatter systems suffer from limited communication range, low data rates, and high sensitivity to multipath fading and indoor attenuation [3].

These limitations remain a major obstacle to the practical deployment of large-scale battery-free sensor networks. The backscatter communication architecture considered in figure 2.2 follows the model presented in *Proceedings of the IEEE* [15]

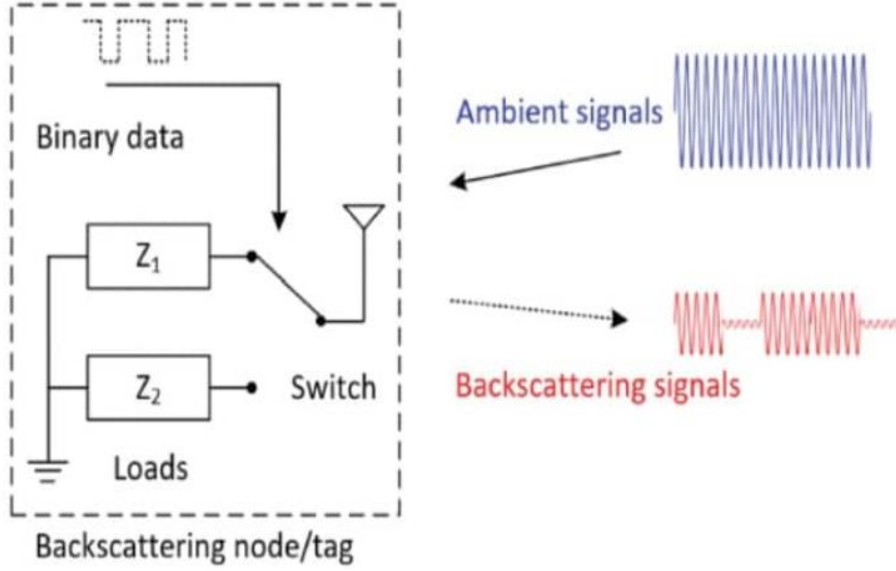


Figure 2.2 Diagram of a backscatter communication tag

2.4. RIS-Assisted Battery-Free and Backscatter Systems

To address the weak signal strength inherent in backscatter communication, recent research has explored integrating RIS with battery-free systems. RIS can be used to redirect and focus ambient RF energy toward backscatter devices, enhancing both energy harvesting and communication reliability [10].

Existing studies primarily rely on analytical models and simulation-based evaluations, often assuming ideal channel knowledge and continuous phase control [11]. While results indicate substantial performance gains, practical challenges such as hardware limitations, discrete phase shifts, control overhead, and dynamic indoor environments are frequently neglected. Experimental validation of RIS-assisted backscatter systems, particularly in realistic indoor scenarios, remains scarce.

2.5. Artificial Intelligence for RIS Optimization

The performance of RIS-assisted systems strongly depends on the configuration of the reflecting elements. Conventional optimization methods typically require accurate channel state information, which is difficult to obtain in dynamic indoor environments.

To overcome this challenge, artificial intelligence (AI) techniques such as machine learning and reinforcement learning have been proposed for adaptive RIS control [9].

Prior work has shown that AI-based approaches can optimize RIS configurations using simple performance feedback metrics, without explicit channel estimation [12]. However, most existing studies focus on improving throughput or coverage for active, battery-powered users. The application of lightweight AI algorithms to RIS-assisted battery-free or backscatter systems remains largely unexplored, especially in experimental and low-complexity settings.

2.6. Related Works

Wireless communication systems are continuously evolving to support the growing demand for energy-efficient and reliable connectivity in Internet of Things (IoT) environments. Recent research has focused on technologies such as ambient backscatter communication, Reconfigurable Intelligent Surfaces (RIS), and artificial intelligence (AI)-based optimization to enable sustainable and low-power wireless networks. These technologies aim to enhance communication reliability, improve signal propagation, and extend the operational lifetime of wireless sensor nodes. This section reviews existing work related to ambient backscatter communication, RIS-assisted wireless communication, RIS-assisted backscatter systems, and AI-based optimization methods.

2.6.1 Ambient Backscatter Communication

Ambient backscatter communication enables low-power devices to transmit information by reflecting existing radio frequency (RF) signals from sources such as Wi-Fi routers, television broadcasts, or cellular base stations. This approach eliminates the need for active RF transmission and allows battery-free devices to operate with extremely low energy consumption.

Early studies demonstrated the feasibility of ambient backscatter communication for short-range wireless communication by modulating and reflecting ambient RF signals [7]. Subsequent research explored methods to improve communication reliability and extend transmission range for battery-free devices [2], [3]. These studies highlighted the potential of ambient backscatter communication for applications such as environmental monitoring, smart sensing, and IoT systems.

Despite these advantages, conventional ambient backscatter systems often suffer from weak signal strength and limited communication range, particularly in indoor environments where signal attenuation and multipath fading are common.

2.6.2 RIS-Assisted Wireless Communication

Reconfigurable Intelligent Surfaces (RIS) have recently emerged as a promising technology for improving wireless communication performance by intelligently controlling electromagnetic wave propagation. An RIS is composed of multiple passive reflecting elements that can dynamically adjust the phase of incident signals to improve signal propagation and communication quality.

Several studies have shown that RIS-assisted communication systems can enhance signal coverage, reduce path loss, and improve spectral efficiency in wireless networks [4], [5]. By manipulating signal reflections, RIS can create favorable propagation paths between transmitters and receivers even in environments with obstacles and signal blockage. Other studies have demonstrated that RIS-assisted systems can outperform conventional relay-based communication while consuming significantly less power [6], [9].

2.6.3 RIS-Assisted Backscatter Communication

Recent research has explored the integration of RIS technology with ambient backscatter communication to address the limitations of conventional battery-free systems. By directing and concentrating RF signals toward sensor nodes, RIS can enhance both energy harvesting efficiency and communication reliability.

Studies have shown that RIS-assisted backscatter communication can significantly improve received signal strength and extend communication range compared to traditional backscatter systems [10]. Intelligent reflection of RF signals can also reduce signal blockage caused by walls and obstacles, making RIS particularly suitable for indoor wireless environments.

2.6.4 AI-Based RIS Optimization

The performance of RIS-assisted communication systems strongly depends on the configuration of the reflecting elements. Traditional optimization techniques often require accurate channel state information, which can be difficult to obtain in complex indoor environments.

To overcome this challenge, recent studies have explored the use of artificial intelligence and machine learning techniques to optimize RIS configurations dynamically. AI-based approaches can learn optimal reflection patterns from system performance feedback, enabling adaptive control of RIS elements in changing environments [11]. These techniques improve signal strength, communication reliability, and system adaptability.

2.6.5 Existing RIS-Backscatter System Implementations

Several recent studies have investigated practical architectures that integrate RIS with backscatter communication. One proposed system utilized RIS to assist IoT devices by reflecting signals from a base station while optimizing reflection phase shifts to improve communication performance. The results showed that increasing the number of RIS elements significantly enhances communication reliability and extends the operational lifetime of IoT devices.

- i. **RIS-enabled joint backscatter and communication framework** where RIS simultaneously supports conventional wireless communication and passive backscatter transmission. By dynamically adjusting reflection patterns, the RIS improves both the primary communication link and the backscatter transmission performance [11].
- ii. **RIScatter architecture** has also been proposed to integrate RIS with battery-free backscatter nodes. In this system, sensor nodes harvest energy from ambient signals while using RIS reflections to transmit information and improve channel conditions [7].
- iii. **Simultaneous Transmitting and Reflecting RIS (STAR-RIS)** more advanced architectures have also been explored, enabling both signal reflection and transmission to enhance communication coverage and reliability [2], [3].

These studies demonstrate that RIS-assisted backscatter systems have significant potential for improving the performance of energy-efficient IoT networks.

2.6.6 Research Gap

Although previous research has demonstrated the potential of RIS, ambient backscatter communication, and AI-based optimization techniques, most existing studies focus primarily on theoretical analysis or simulation-based evaluations. Practical implementations of RIS-assisted battery-free communication systems remain limited, particularly in indoor environments.

Furthermore, many optimization approaches rely on complex algorithms that may not be suitable for real-time deployment in resource-constrained systems. Limited attention has been given to small-scale experimental prototypes that integrate RIS, battery-free communication, and lightweight AI algorithms.

CHAPTER THREE

3. METHODOLOGY

3.1. System Architecture

The system consists of the following main subsystems:

1. Ambient RF Signal Source
2. Battery-Free Backscatter Sensor Node
3. Reconfigurable Intelligent Surface (RIS)
4. Receiver Node
5. AI-Based RIS Control Unit

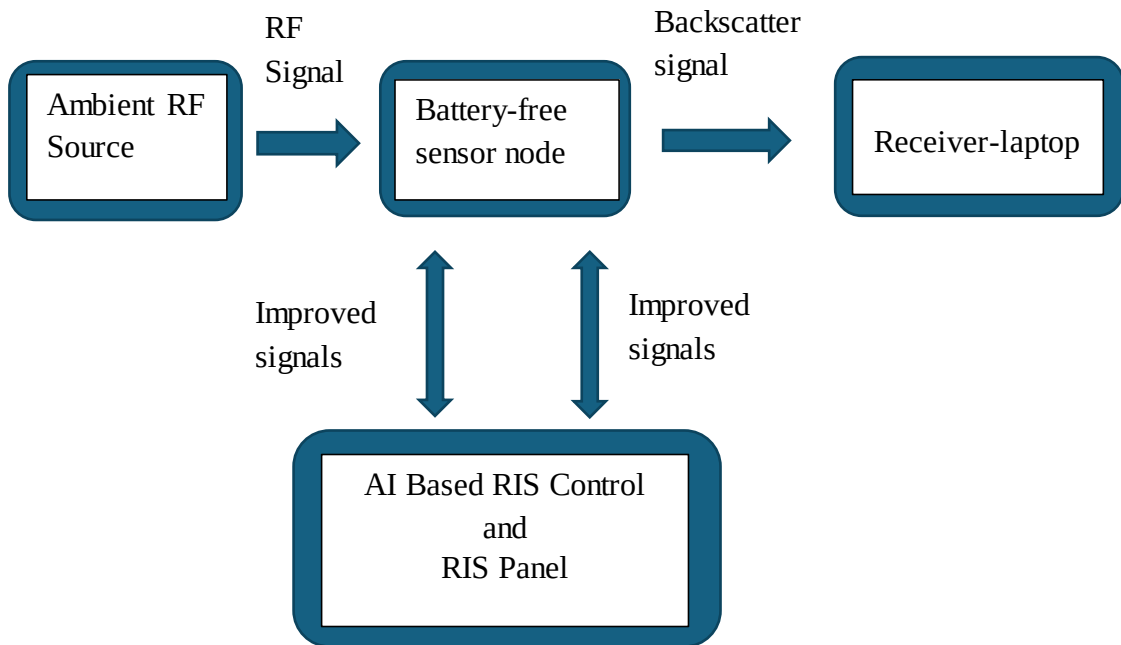


Figure 3.1 Block diagram of the proposed RIS-assisted network.

Each subsystem is described in detail below.

3.1.1 Ambient RF Signal Source

The RF source provides ambient radio frequency signals used for communication and energy harvesting. In this project, a Nokia gateway router acts as the RF signal source generating Wi-Fi signals.

3.1.2 Battery-Free Backscatter Sensor Node

The battery-free sensor node is designed to operate without a conventional power source by utilizing energy harvesting and backscatter communication principles. The methodology is based on minimizing power consumption while enabling data transmission through passive RF techniques. The node is conceptually designed to harvest energy from ambient sources and use this energy to intermittently power sensing and communication operations. Once sufficient energy is available, the sensor measures environmental data (e.g., temperature) and prepares it for transmission.

Data transmission is achieved using a backscatter modulation approach, where the antenna impedance is switched between two states to encode binary information. This switching creates variations in the reflected RF signal, representing digital data ('0' and '1') without generating a new RF signal. A simple switching mechanism using a transistor is employed to control the antenna reflection characteristics. This enables the node to transmit sensed data efficiently with minimal energy consumption.

Overall, this focuses on ultra-low-power operation, energy-efficient sensing, and passive communication, making it suitable for battery-free and energy-constrained wireless systems.

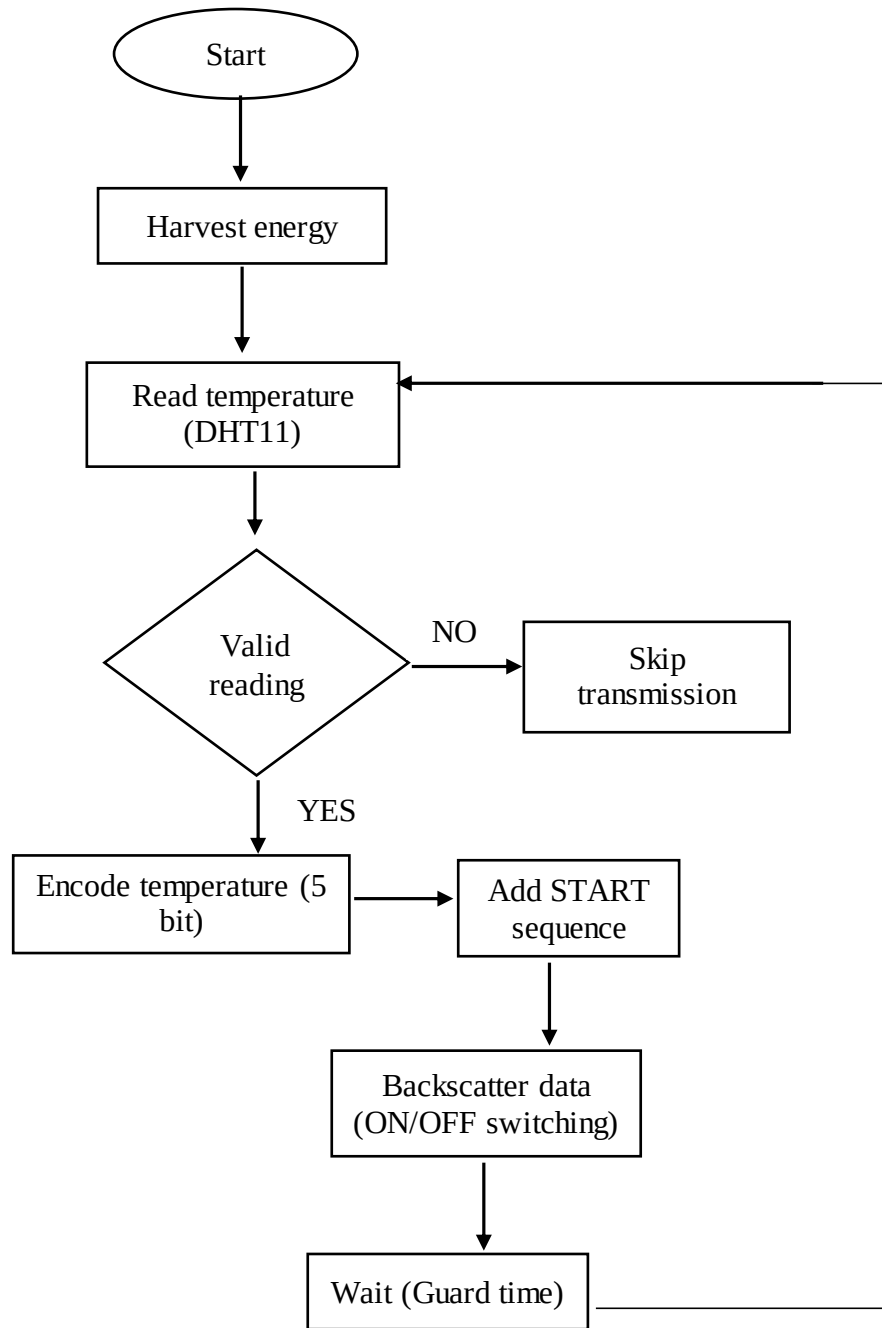


Figure 3.2 Battery-free sensor node operation workflow

3.1.3 Reconfigurable Intelligent Surface (RIS)

The RIS is designed to dynamically control the propagation of RF signals. The RIS enhances energy harvesting and backscatter signal strength by intelligently reflecting RF waves. The RIS consists of multiple reflective elements whose states can be adjusted electronically, each RIS element can independently adjust the strength of the reflected signal. This changes how Wi-Fi signal bounces and is called binary amplitude

control RIS. By configuring the phase shifts across the surface with the use of PIN diodes, the RIS directs electromagnetic waves toward specific locations, effectively shaping the indoor radio environment. The RIS is controlled using a microcontroller-based controller an ESP32, which modifies the reflection patterns according to instructions received from the AI optimization system.

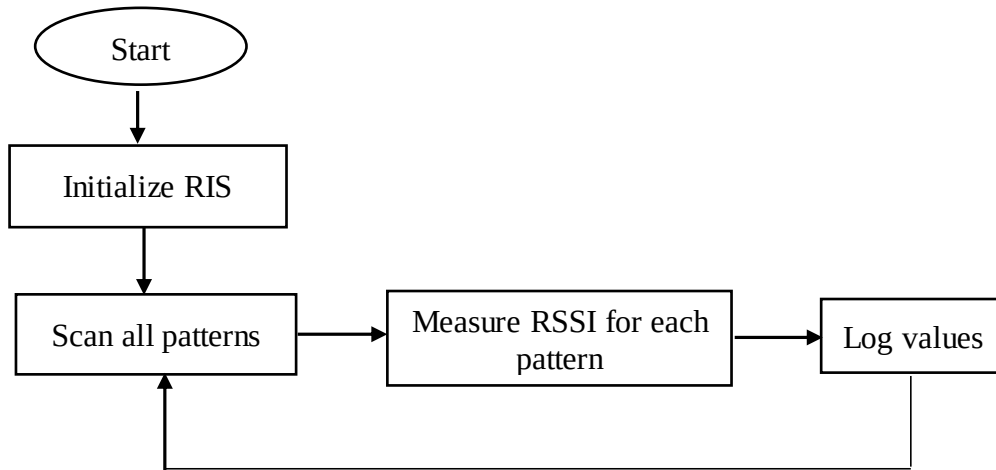


Figure 3.3 RIS operation workflow

3.1.4 Receiver Node

The receiver node is designed to detect and decode temperature data transmitted through backscatter communication using variations in signal strength. It is based on leveraging RSSI (Received Signal Strength Indicator) as the primary metric for interpreting modulated signals from the battery-free sensor node. The receiver continuously monitors these variations to extract meaningful information. The receiver node operates by first performing Wi-Fi signal scanning to detect signals from the backscatter sensor node. It continuously samples RSSI values, which vary due to the sensor's data modulation.

These RSSI variations are processed using a threshold-based detection method, where signal levels are mapped into binary states ('1' and '0'). This allows conversion

of signal strength changes into a digital bit stream. The reconstructed bit stream is then decoded to recover the transmitted temperature data. The decoded values are finally sent to an external interface for visualization.

Overall, this relies on simple RSSI monitoring and threshold decoding, enabling real-time, low-complexity extraction of sensor data in a backscatter communication system hence implementing a functional receiver for backscatter communication systems.

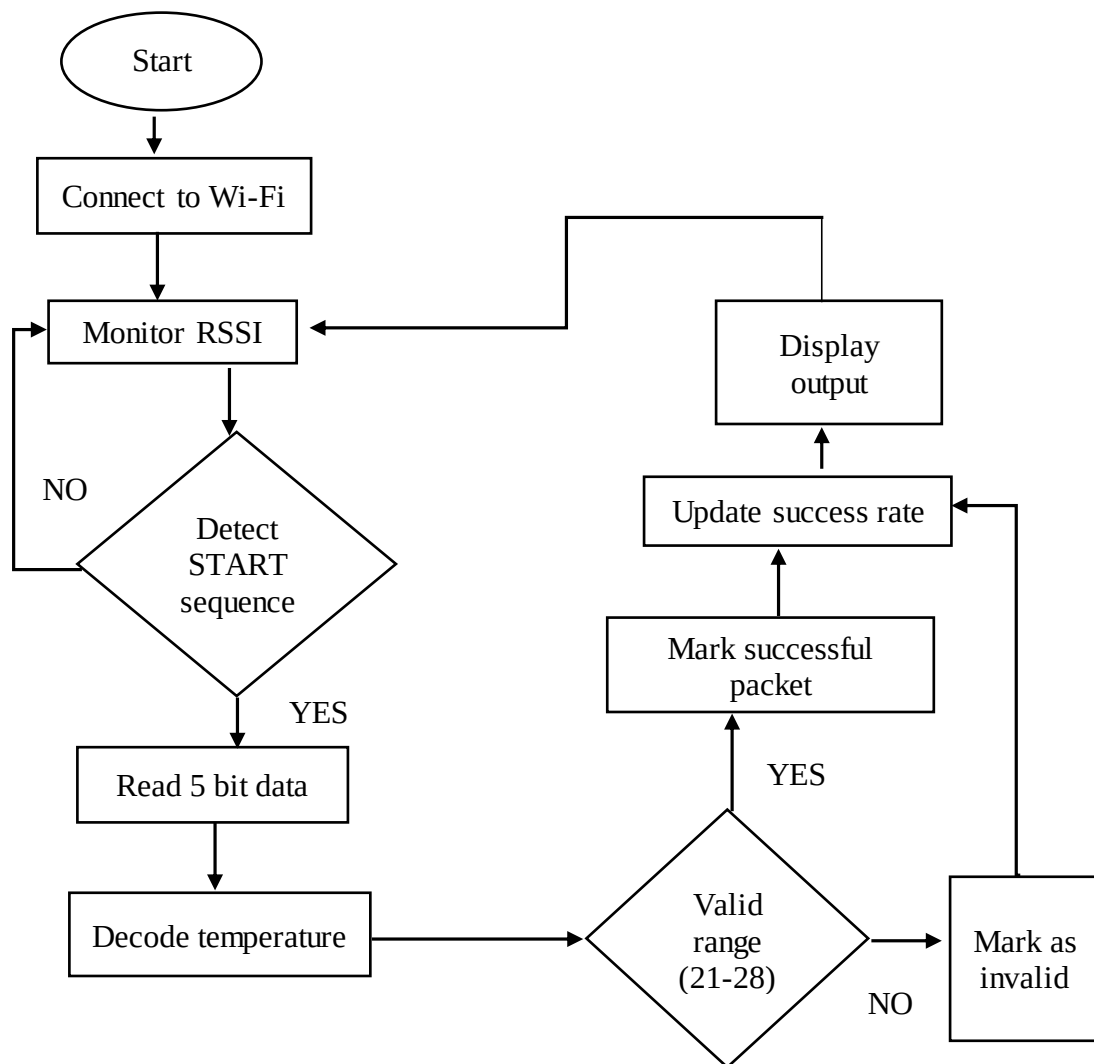


Figure 3.4 Receiver node operation workflow

3.1.5 Intelligent-Based RIS Control Unit

The RIS control unit is implemented using an ESP32 microcontroller, which executes the system's intelligent pattern selection and optimization logic. The ESP32 manages the switching states of the RIS elements and coordinates different predefined reflection configurations.

An intelligent pattern selection mechanism is implemented where the controller cycles through multiple predefined RIS configurations and evaluates their performance based on the Received Signal Strength Indicator (RSSI) measured at the receiver. Each configuration is tested, and the corresponding RSSI value is recorded for comparison.

The configuration that produces the highest RSSI is selected as the optimal reflection pattern. This enables adaptive optimization of signal propagation in real time, allowing the RIS to enhance communication performance by dynamically selecting the most effective reflection state without requiring complex external processing systems or additional computational hardware.

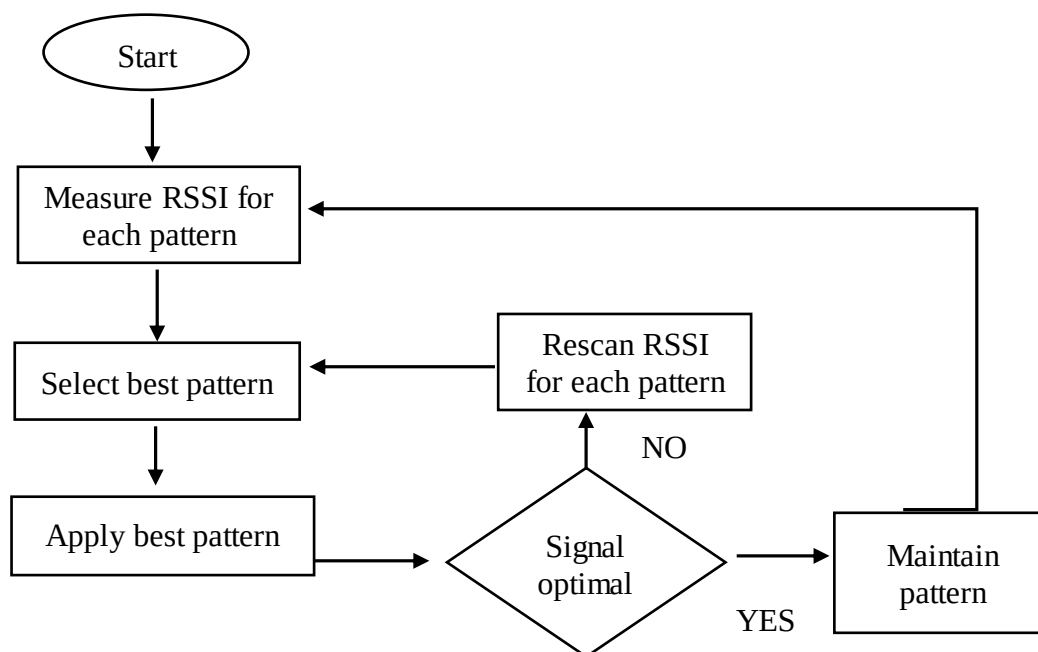


Figure 3.5 Intelligent RIS operation workflow

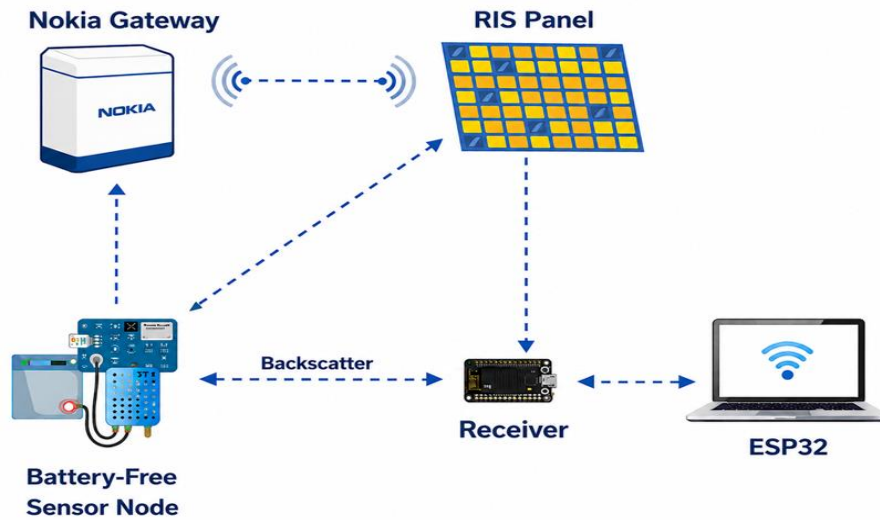


Figure 3.6 System Architecture

3.2. System Operation Workflow

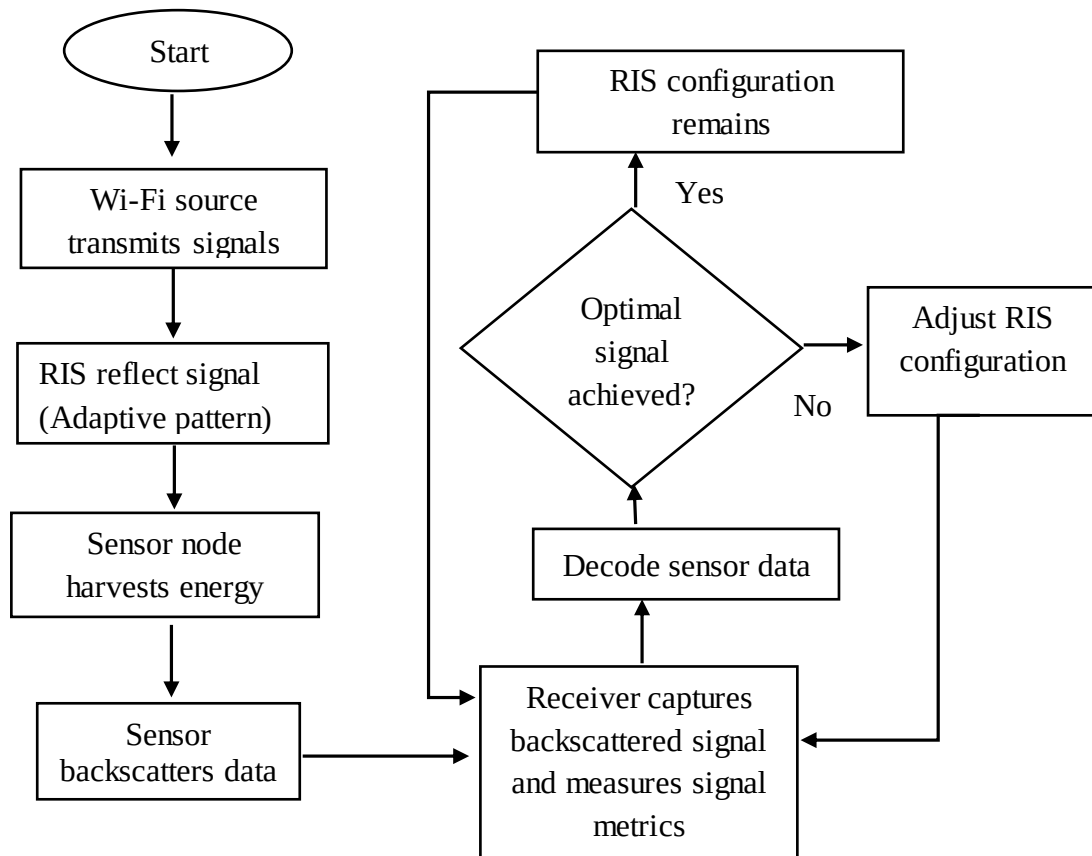


Figure 3.7 Operation workflow

3.3. Requirements

3.3.1 Hardware Components

a) Signal Source

The signal source used in this project is a Nokia gateway router, which generates ambient Wi-Fi signals that serve as the RF energy source for backscatter communication.

b) Reconfigurable Intelligent Surface

The RIS subsystem consists of the following hardware components:

- RIS panel, 3×3 reflective surface structure made using aluminium foil elements mounted on a flat substrate
- PIN diodes (2 active elements) used to enable electronic switching of selected reflective elements
- ESP32 microcontroller, responsible for controlling the switching states of the PIN diodes and managing reflection patterns
- Resistors and biasing circuit components, used to safely control current flow through the PIN diodes
- Power supply (USB) used to power the ESP32 control unit
- Breadboard provides a platform for circuit prototyping and integration

c) Sensor Node

The battery-free sensor node is designed using low-power components to support energy harvesting and backscatter communication. The required hardware includes:

- Solar panel – Serves as the primary energy source for harvesting ambient light energy

- Capacitors – Stores harvested energy and supplies power to the sensor during operation
- Environmental sensor (temperature) – Measures the physical parameter to be transmitted
- Copper wire antenna – Used for receiving and reflecting RF signals for backscatter communication
- NPN transistor – Acts as a switching element to modulate antenna impedance for data encoding
- Schottky diode – Included in the design for efficient rectification (initial RF harvesting concept)
- Resistors – Used for biasing and current control within the circuit
- Breadboard – Provides a platform for circuit prototyping and integration

d) Receiver

The receiver node is built using a minimal hardware setup to ensure low cost and simplicity. The main component is an ESP32 microcontroller, which is responsible for scanning and receiving backscatter signals through its Wi-Fi interface. It is powered and programmed through a USB connection to a laptop, which also enables serial communication between the device and the processing interface.

A laptop is used as the external processing and display unit. It runs a serial monitor or terminal application to receive and visualize decoded temperature data transmitted from the ESP32.

3.3.2 Software Components

The system software components are used for programming, configuration control, signal monitoring, and performance evaluation. The main software tools include:

- Arduino IDE – Used for programming the ESP32 microcontroller responsible for controlling the RIS, sensor node, and receiver operations.
- PuTTY– Used to display real-time RSSI values of the RIS transmitted from the ESP32 via bluetooth.
- MATLAB – Used for simulation and analysis of signal behavior, including evaluation of RIS performance and pattern comparisons.

The software environment enables system configuration, data acquisition, and performance evaluation of the RIS-based backscatter communication system.

3.4. Simulation of 6G-Inspired Communication Environment

The simulation component is incorporated to complement the experimental prototype by evaluating a 6G-inspired indoor communication environment under controlled conditions. It is designed to model wireless signal propagation in relevant frequency bands, including sub-6 GHz and millimetre-wave ranges, while assessing the impact of Reconfigurable Intelligent Surface (RIS)-assisted reflections on system performance metrics such as received signal strength (RSSI) and signal-to-noise ratio (SNR). The simulation incorporated high frequency 28GHz communication which exhibits significant path loss and sensitivity to obstacles. The simulation is implemented in MATLAB using custom-developed models for indoor wireless propagation, RIS behaviour, and backscatter-inspired communication. The environment consists of an RF source, a sensor node which is represented as implicit

link behaviour, a programmable RIS, and a receiver, with fixed distances corresponding to the physical setup.

Wireless signal propagation is modelled using a simplified indoor path loss model incorporating wall attenuation to account for signal degradation due to distance and obstruction. Environmental effects such as multipath fading and shadowing are approximated using random channel variations to reflect realistic indoor conditions. The RIS is modelled as a configurable reflecting system capable of altering its reflection characteristics through different discrete configurations. These configurations are evaluated under three scenarios: without RIS, with non-optimized RIS, and with adaptive intelligent RIS control. In the intelligent mode, the RIS dynamically selects the configuration that maximizes the received signal strength under varying channel conditions.

System performance is evaluated using key metrics, including received signal strength (RSSI), signal-to-noise ratio (SNR), and link reliability, where SNR is derived from the received power relative to the noise floor, and reliability is defined as the proportion of successful communication instances exceeding a predefined threshold. This simulation framework provides a basis for validating the effectiveness of RIS-assisted and adaptive RIS-controlled communication in enhancing backscatter system performance in indoor environments.

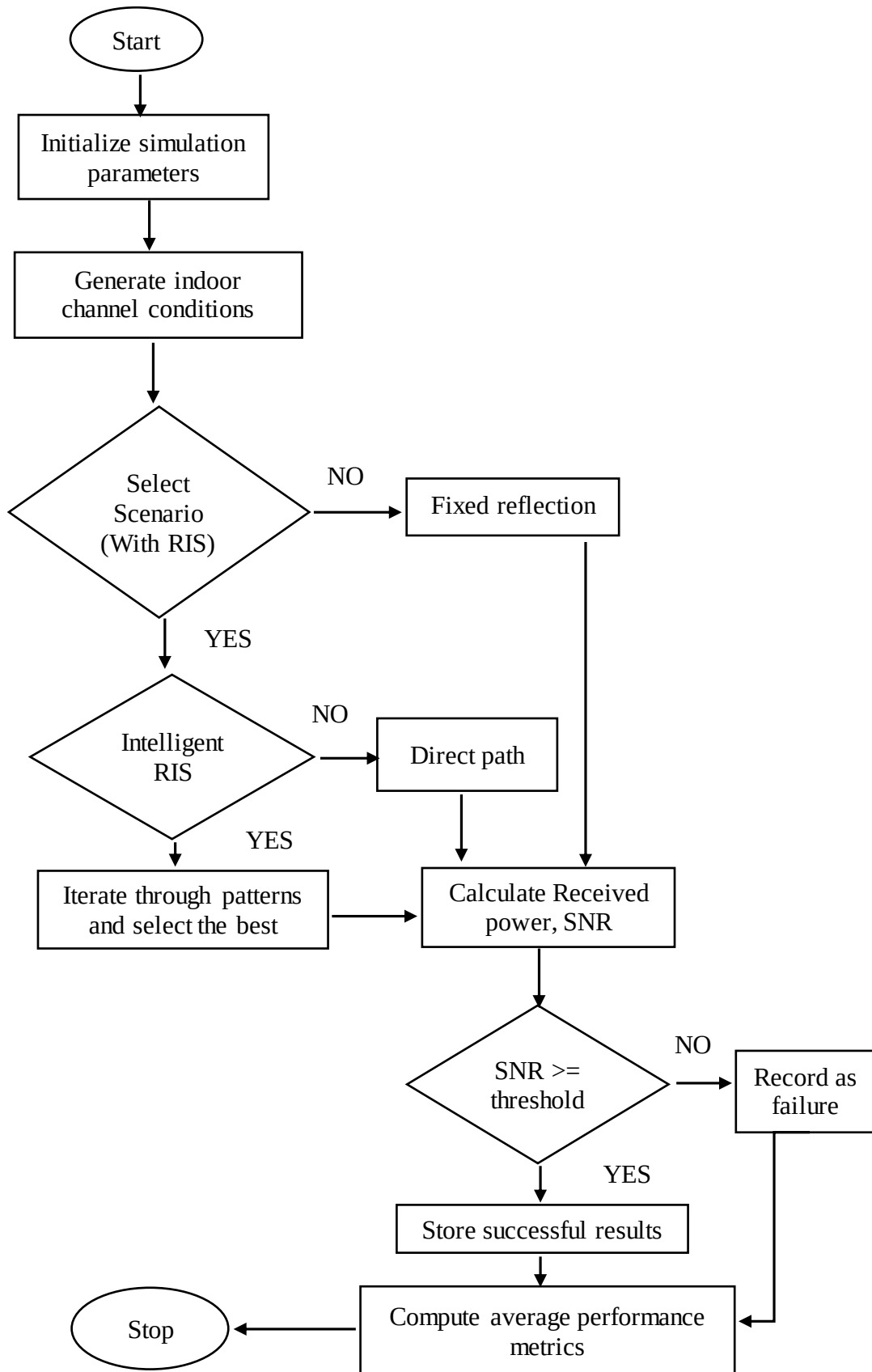


Figure 3.8 Simulation workflow

CHAPTER FOUR

4. IMPLEMENTATION

4.1. Hardware Implementation

4.1.1. Battery-Free Backscatter Sensor Node

The battery-free sensor node was implemented using a solar-based energy harvesting approach, replacing the initially proposed RF energy harvesting method due to practical limitations in achieving sufficient power from ambient RF sources. The solar panel harvests energy from ambient light and stores it in a capacitor, which functions as an energy buffer. Once the stored energy reaches a usable level, the temperature sensor is powered briefly to take a measurement.

Backscatter communication is implemented by modulating the antenna's reflection characteristics using an NPN transistor. The transistor switches between ON and OFF states, altering the antenna impedance and creating variations in the reflected RF signal. These variations encode the sensed data into binary form. The encoded signal is then reflected toward the receiver using ambient RF signals from an external source. This eliminates the need for active RF transmission, significantly reducing power consumption.

The final implementation demonstrates a functional battery-free sensing system powered by solar energy, combined with backscatter-based data transmission, achieving low-power operation suitable for energy-constrained environments.

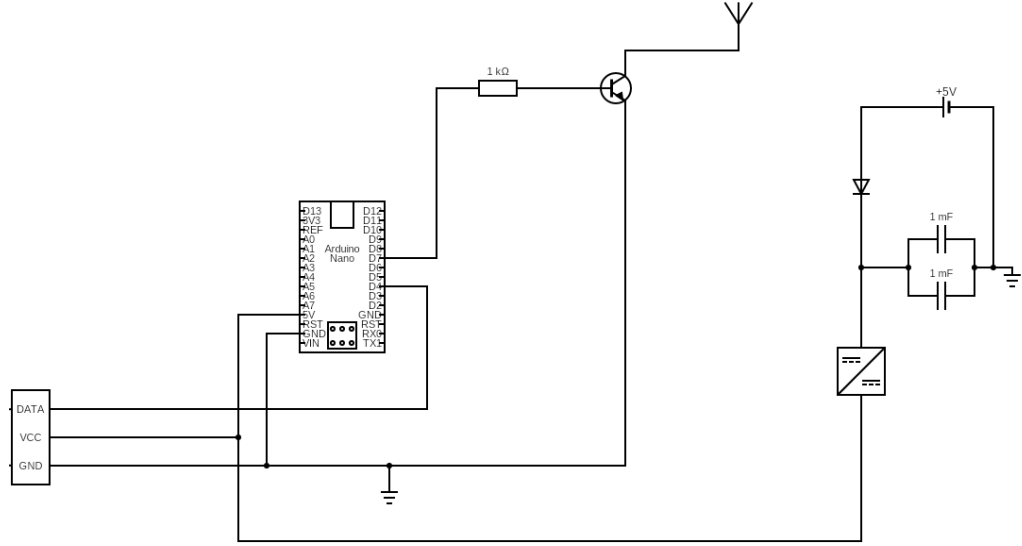


Figure 4.1 Circuit representation of battery-free backscatter sensor node

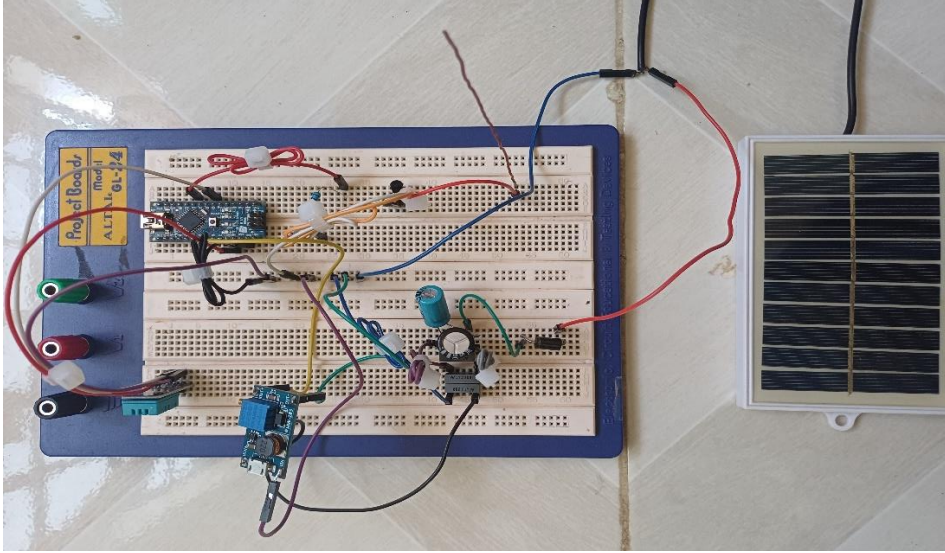


Figure 4.2 Physical prototype of battery-free sensor node

4.1.2. Reconfigurable Intelligent Surface (RIS)

The Reconfigurable Intelligent Surface (RIS) was implemented as a low-cost, programmable reflecting structure designed to manipulate incident RF signals. Due to practical constraints, a simplified prototype was developed using a 3×3 array of aluminium foil elements arranged on a flat substrate. These elements act as passive reflectors that influence signal propagation through reflection and scattering.

To introduce reconfigurability, two of the nine elements were integrated with PIN diodes, enabling dynamic control of their reflection properties, while the remaining elements functioned as static reflectors. The PIN diodes operate as RF switches, toggling between ON (strong reflection) and OFF (reduced reflection) states through forward and reverse biasing. The RIS was strategically positioned between the RF source (Nokia gateway) and the receiver to enhance signal strength.

Control of the RIS was achieved using an ESP32 microcontroller, which regulates the PIN diode states. This allows selective activation of elements, dynamic adjustment of reflection patterns, and testing of multiple configurations to influence signal propagation. Although limited in scale, this demonstrates intelligent surface behavior by enabling the RIS to adapt its response based on control inputs.

A bluetooth-based interface was implemented using the ESP32 to support real-time monitoring and control. In manual mode, the RIS cycles through predefined configurations while transmitting corresponding RSSI values. In intelligent mode, the system evaluates all configurations and automatically selects the one that maximizes RSSI at the receiver. The selected configuration and performance metrics are continuously transmitted to an external device such as a laptop or smartphone.

Overall, the system integrates the RIS, ESP32 control unit, and Bluetooth communication into a unified architecture, demonstrating a practical and scalable approach to adaptive signal enhancement aligned with emerging 6G communication concepts.

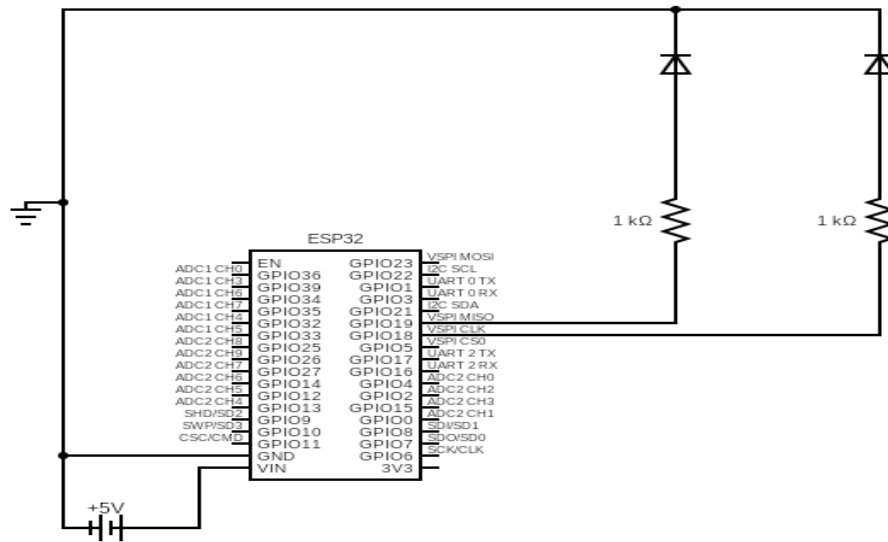


Figure 4.3 Circuit implementation of RIS

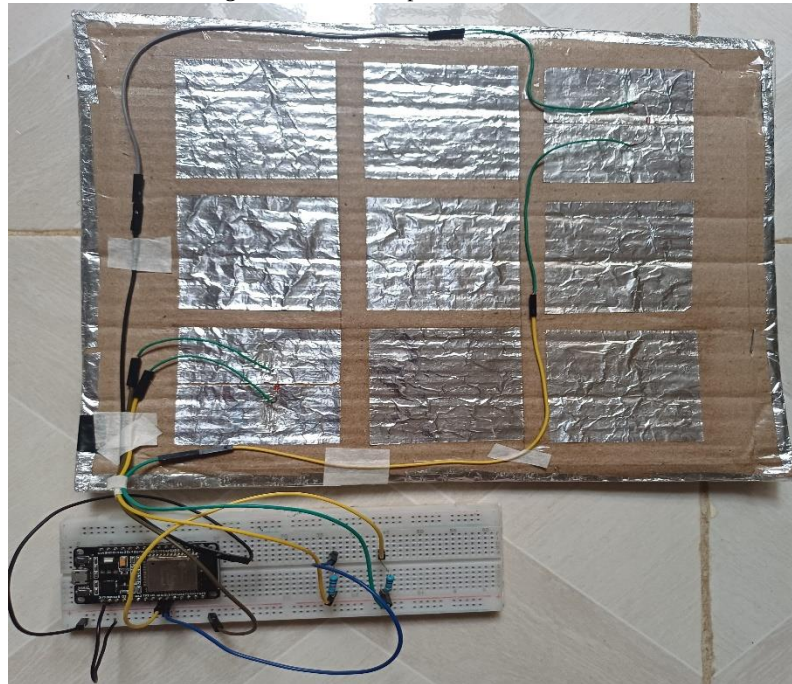


Figure 4.4 Physical prototype of the RIS

4.1.3. ReceiverNode

The receiver node consists of a single ESP32 microcontroller connected to a laptop via a USB cable, where the laptop serves only as a display interface. The ESP32 is powered directly through the USB connection, which also enables serial communication.

The ESP32 operates in Wi-Fi scanning mode to capture backscatter signals from the battery-free sensor node. Instead of decoding standard packets, it relies on RSSI (Received Signal Strength Indicator) variations, which carry the encoded temperature data. These variations are continuously sampled and processed in real time.

Data decoding is achieved by interpreting RSSI fluctuations using predefined thresholds to distinguish binary states. The resulting bit stream is reconstructed and converted into integer temperature values within the ESP32 firmware.

Once decoded, the temperature data is transmitted to the laptop via serial communication (UART over USB) and displayed in real time using a serial monitor. The laptop performs no processing and functions solely as a visualization tool.

Overall, the receiver node captures backscatter signals, decodes temperature data using RSSI-based processing, and outputs the results to a laptop, providing a simple and effective low-complexity implementation.

4.2. Software Implementation

4.2.1. Battery-Free Backscatter Sensor Node

a) Sensor Initialization and Setup

This section initializes the temperature sensor and configures the switching pin used for backscatter modulation. The DHT11 sensor is initialized for temperature measurement, and the switching pin is configured as an output to control backscatter modulation.

```
1  #include <DHT.h>
2
3  #define DHTPIN 4
4  #define DHTTYPE DHT11
5  #define SWITCH_PIN 7
6
7  DHT dht(DHTPIN, DHTTYPE);
8
34 void setup() {
35   Serial.begin(9600);
36   pinMode(SWITCH_PIN, OUTPUT);
37   dht.begin();
38   delay(2000);
39 }
```

Figure 4.5 Sensor node initialization

b) Bit Transmission Function

This function controls how individual bits are transmitted using antenna switching

```
void sendBit(int bit) {  
    if (bit == 1) {  
        digitalWrite(SWITCH_PIN, HIGH);  
    } else {  
        digitalWrite(SWITCH_PIN, LOW);  
    }  
    delay(600);  
}
```

Figure 4.6 Bit transmission function

Each bit is transmitted by toggling the switching pin. HIGH and LOW states correspond to different antenna impedance states, encoding binary '1' and '0'.

c) Data Encoding and Transmission

This section defines how temperature data is structured and transmitted.

```
sendBit(1);  
sendBit(1);  
sendBit(1);  
sendBit(1);  
// -----  
// Send 5-bit temperature  
// -----  
for (int i = 4; i >= 0; i--) {  
    int bitValue = (temp >> i) & 1;  
    sendBit(bitValue);  
}  
// End transmission  
digitalWrite(SWITCH_PIN, LOW);  
}
```

Figure 4.7 Data encoding

A start sequence (1111) is first transmitted to signal the beginning of data. The temperature value is then encoded into a 5-bit binary format and sent sequentially.

d) Sensor Reading and Transmission Loop

This section continuously reads temperature data and transmits it.

```
void loop() {  
    delay(2000);  
    int temp = dht.readTemperature();  
    // Invalid sensor reading  
    if (isnan(temp)) {  
        Serial.println("Sensor read failed");  
        return;  
    }  
}
```

Figure 4.8 Sensor reading and transmission

The sensor reads the temperature and checks for errors. If the reading fails, a default value is used. The data is then transmitted using backscatter modulation in regular intervals.

4.2.2. Reconfigurable Intelligent Surface (RIS)

a) Wi-Fi and Bluetooth Initialization

This section initializes Wi-Fi connectivity and Bluetooth communication for control and monitoring. The ESP32 connects to a Wi-Fi network for RSSI measurement and initializes Bluetooth communication to allow remote control and monitoring via external devices.

```
1  #include <WiFi.h>
2  #include "BluetoothSerial.h"
3
4  BluetoothSerial SerialBT;
5
6  // WiFi credentials
7  const char* ssid = "NOKIA-F0A654C57ACB";
8  const char* password = "4LRL8kJ3E2";
119 void setup() {
120     Serial.begin(115200);
121     SerialBT.begin("ESP32_RIS");
122
123     pinMode(d1, OUTPUT);
124     pinMode(d2, OUTPUT);
125
126     Serial.println("Connecting to WiFi...");
127     SerialBT.println("Connecting to WiFi...");
128
129     WiFi.begin(ssid, password);
130
131     while (WiFi.status() != WL_CONNECTED) {
132         delay(500);
133         Serial.print(".");
134         SerialBT.print(".");
135     }
136
137     Serial.println("\nConnected to WiFi");
138     SerialBT.println("\nConnected to WiFi");
```

Figure 4.9 RIS Wi-Fi and bluetooth initialization

b) RIS Pattern Configuration

This function controls the switching states of the RIS elements using PIN diodes.

```

24 void setPattern(int pattern) {
25     switch(pattern) {
26         case 0:
27             digitalWrite(d1, LOW);
28             digitalWrite(d2, LOW);
29             break;
30         case 1:
31             digitalWrite(d1, HIGH);
32             digitalWrite(d2, LOW);
33             break;
34         case 2:
35             digitalWrite(d1, LOW);
36             digitalWrite(d2, HIGH);
37             break;
38         case 3:
39             digitalWrite(d1, HIGH);
40             digitalWrite(d2, HIGH);
41             break;
42     }
43 }

```

Figure 4.10 RIS pattern configuration

Different combinations of HIGH and LOW signals control the ON/OFF states of the PIN diodes, creating multiple RIS reflection patterns.

c) RSSI Measurement

This function calculates the average RSSI value for a given RIS configuration. Multiple RSSI samples are taken and averaged to reduce noise and improve measurement reliability.

```

48 int getAverageRSSI() {
49     int total = 0;
50     const int samples = 5;
51     for (int i = 0; i < samples; i++) {
52         total += WiFi.RSSI();
53         delay(200);
54     }
55     return total / samples;
56 }

```

Figure 4.11 RSSI measurement

d) Intelligent Pattern Selection

This section implements the intelligent RIS behavior by selecting the best-performing pattern. The system cycles through all possible RIS configurations, measures their RSSI performance, and selects the pattern that provides the strongest signal.

```

61 void findBestPattern() {
62     bestRSSI = -1000;
63
64     for (int p = 0; p < 4; p++) {
65         setPattern(p);
66         delay(1000);
67
68         int rssi = getAverageRSSI();
69
70         Serial.print("Pattern ");
71         Serial.print(p + 1);
72         Serial.print(" RSSI: ");
73         Serial.println(rssi);
74
75         SerialBT.print("Pattern ");
76         SerialBT.print(p + 1);
77         SerialBT.print(" RSSI: ");
78         SerialBT.println(rssi);
79
80         if (rssi > bestRSSI) {
81             bestRSSI = rssi;
82             bestPattern = p;
83         }
84     }

```

Figure 4.12 Intelligent pattern selection

e) Manual (Non-Intelligent) Mode

This function continuously displays RSSI values for all patterns without optimization. This mode allows monitoring of all RIS configurations without selecting the best one, useful for comparison and analysis.

```

98 void displayPatternsRSSIContinuous() {
99     for (int p = 0; p < 4; p++) {
100         setPattern(p);
101         delay(1000); // stabilize
102         int rssi = getAverageRSSI();
103
104         Serial.print("Pattern ");
105         Serial.print(p + 1);
106         Serial.print(" RSSI: ");
107         Serial.println(rssi);
108
109         SerialBT.print("Pattern ");
110         SerialBT.print(p + 1);
111         SerialBT.print(" RSSI: ");
112         SerialBT.println(rssi);
113     }
114 }

```

Figure 4.13 Non-Intelligent RIS

f) Control Logic (Bluetooth Interface)

This section allows switching between intelligent and manual modes. The system receives commands via Bluetooth to enable or disable intelligent RIS operation. In intelligent mode, the best pattern is selected automatically, while in manual mode, all patterns are continuously evaluated and displayed.

```

18 // Flag to control whether Intelligent RIS is active
19 bool intelligentRISActive = false;

147 void loop() {
148 // Check for PuTTY/Bluetooth input
149 if (SerialBT.available()) {
150 String cmd = SerialBT.readStringUntil('\n');
151 cmd.trim();
152
153 if (cmd.equalsIgnoreCase("ON")) {
154 intelligentRISActive = true;
155 Serial.println("Intelligent RIS ACTIVATED");
156 SerialBT.println("Intelligent RIS ACTIVATED");
157 } else if (cmd.equalsIgnoreCase("OFF")) {
158 intelligentRISActive = false;
159 Serial.println("Intelligent RIS DEACTIVATED: Displaying all patterns RSSI continuously");
160 SerialBT.println("Intelligent RIS DEACTIVATED: Displaying all patterns RSSI continuously");
161 }
162 }
163
164 if (intelligentRISActive) {
165 // Run Intelligent RIS
166 findBestPattern();
167 } else {
168 // Continuously show RSSI for all patterns without intelligence
169 displayPatternsRSSIContinuous();
170 }
171 }
---
```

Figure 4.14 RIS Control logic

4.2.3. ReceiverNode

a) Wi-Fi configuration and initialization

This section initializes serial communication and connects the ESP32 to the Wi-Fi network, enabling RSSI-based signal monitoring.

```

1 #include <WiFi.h>
2 const char* ssid = "NOKIA-F0A654C57ACB";
3 const char* password = "4LRL8kJ3E2";
4
5 int threshold = -32;
6 int bitDuration = 1000;
7 // -----
8 int totalPackets = 0;
9 int successfulPackets = 0;
10 // -----
11 int readBit() {
12 int rssi = WiFi.RSSI();
13 return (rssi > threshold) ? 1 : 0;
14 }
15 // -----
16 void setup() {
17 Serial.begin(115200);
18
19 WiFi.begin(ssid, password);
20
21 while (WiFi.status() != WL_CONNECTED) {
22 delay(500);
23 Serial.print(".");
24 }
25
26 Serial.println("\nReceiver Ready");
27 }
```

Figure 4.15 Receiver initialization

The ESP32 connects to a specified Wi-Fi network and continuously checks connection status until successful. Once connected, it is ready to begin RSSI-based signal detection for backscatter communication. RSSI values are compared against a predefined threshold. Values above the threshold represent binary '1', while those below represent '0'.

b) Start Bit Detection

This section detects the beginning of a valid transmission using a predefined pattern of consecutive '1's.

```
bool detectStartFrame() {
    int pattern[4];
    for (int i = 0; i < 4; i++) {
        delay(SAMPLE_DELAY);
        int rssi = WiFi.RSSI();
        int diff = rssi - baseline;
        // Ignore weak RSSI fluctuations
        if (abs(diff) < thresholdStart) {
            return false;
        }
    }
}
```

Figure 4.16 Receiver start bit detection

The receiver continuously samples RSSI and converts it into binary values. A valid transmission is identified when three consecutive '1's are detected, indicating the start of a data packet.

c) Data Bit Reading

Once the start sequence is detected, the receiver reads the incoming data bits.

```
42 // Read 5-bit data
43 int value = 0;
44
45 for (int i = 0; i < 5; i++) {
46     int bit = readBit();
47     value = (value << 1) | bit;
48     delay(bitDuration);
49 }
50
51 }
```

Figure 4.17 Receiver data bit reading

Five bits are sequentially read and combined using bitwise shifting to reconstruct the transmitted temperature value.

d) Data Validation

```
56 // VALIDATION (21-28 range)
57 // -----
58 bool isValid = (value >= 21 && value <= 28);
59
60 if (isValid) {
61     successfulPackets++;
62 }
```

Figure 4.18 Data validation

The decoded value is validated against an expected temperature range to filter out noise and incorrect readings.

e) Output and Performance Metrics

```
67 // OUTPUT
68 // -----
69 if (isValid) {
70
71     Serial.print("Received: Temp=");
72     Serial.print(value);
73     Serial.print("°C | RSSI=");
74     Serial.print(rssi);
75     Serial.print(" dBm | Success rate=");
76     Serial.print(successRate);
77     Serial.println("%");
78
79 } else {
80
81     Serial.print("Invalid value=");
82     Serial.print(value);
83     Serial.print(" | RSSI=");
84     Serial.print(rssi);
85     Serial.print(" dBm | Success rate=");
86     Serial.print(successRate);
87     Serial.println("%");
88 }
89
90 Serial.println("-----");
```

Figure 4.19 Receiver output

Valid readings are counted to compute a success rate. The ESP32 then outputs the decoded temperature, RSSI, and system performance metrics via serial communication.

CHAPTER FIVE

5. RESULTS AND DISCUSSION

This chapter presents the evaluation of the developed intelligent RIS-assisted battery-free sensor network. The results are analyzed based on the system's ability to achieve the stated objectives, including reliable data transmission, energy-efficient operation, and improved signal propagation using RIS.

The system was evaluated in an indoor environment using a nokia gateway as the RF source. The battery-free sensor node transmitted temperature data via backscatter, while the ESP32 receiver decoded the signal using RSSI-based processing.

Three operating conditions were considered:

- Without RIS
- With RIS (static configuration)
- With Intelligent RIS (adaptive configuration)

Performance was evaluated using:

- RSSI (dBm) – signal strength
- Packet Success Rate (%) – transmission reliability
- Decoded Data Accuracy – correctness of received temperature values

System Responsiveness (Latency Observation) – time between transmission and reception

5.1. Hardware Testing

The system components were tested individually:

a) Battery-Free Sensor Node

The solar-powered node successfully operated without a battery and was able to periodically sense and transmit temperature data. This confirms the feasibility of low-power, energy-harvesting-based sensing. The system successfully implemented a battery-free sensor node powered by solar energy, capable of sensing and transmitting temperature data. This meets the objective of low-energy operation.

b) Receiver Node

The ESP32 reliably detected RSSI variations and decoded transmitted data. Start sequence detection and bit reconstruction were consistent.

c) RIS

Switching RIS configurations resulted in observable changes in RSSI, confirming its ability to manipulate signal propagation. The RIS significantly improved communication reliability, particularly in intelligent mode as shown in the system performance evaluation.

5.2. Software Testing

The software modules were evaluated for correctness and reliability:

- ✓ Start sequence detection correctly identified packet boundaries
- ✓ RSSI thresholding successfully converted signal variations into binary data
- ✓ Data decoding produced valid temperature values within expected ranges
- ✓ Validation logic filtered incorrect readings
- ✓ Bluetooth interface enabled switching between intelligent and manual RIS modes
- ✓ The software demonstrated stable real-time performance across all test scenarios.

5.3. System Performance Evaluation

5.3.1. Without RIS

Observations:

- Low and unstable RSSI values
- Frequent decoding errors
- High number of invalid packets
- Poor transmission reliability

Without RIS, signal propagation is limited and highly affected by environmental conditions. The weak backscatter signal leads to unreliable communication, demonstrating the need for signal enhancement mechanisms.

```
Receiver Ready
Invalid value=30 | RSSI=-44 dBm | Success rate=0.00%
-----
Received: Temp=29°C | RSSI=-42 dBm | Success rate=50.00%
-----
Received: Temp=27°C | RSSI=-42 dBm | Success rate=66.67%
-----
Invalid value=30 | RSSI=-45 dBm | Success rate=50.00%
-----
Invalid value=31 | RSSI=-42 dBm | Success rate=40.00%
-----
Received: Temp=27°C | RSSI=-43 dBm | Success rate=50.00%
-----
Received: Temp=27°C | RSSI=-39 dBm | Success rate=57.14%
-----
Received: Temp=23°C | RSSI=-43 dBm | Success rate=62.50%
-----
Invalid value=31 | RSSI=-43 dBm | Success rate=55.56%
-----
Invalid value=30 | RSSI=-41 dBm | Success rate=50.00%
-----
Invalid value=31 | RSSI=-45 dBm | Success rate=45.45%
-----
```

Figure 5.1 Receiver output without RIS

5.3.2. With RIS (Static Configuration)

Observations:

- Improved RSSI compared to no-RIS case
- Moderate increase in signal stability
- Improved but inconsistent packet success rate

- Performance dependent on selected pattern

The RIS improves signal reflection and enhances communication. However, static configurations do not guarantee optimal performance, as different patterns produce varying results.

```
Receiver Ready  
Received: Temp=22°C | RSSI=-37 dBm | Success rate=100.00%  
-----  
Received: Temp=27°C | RSSI=-41 dBm | Success rate=100.00%  
-----  
Received: Temp=23°C | RSSI=-37 dBm | Success rate=100.00%  
-----  
XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX  
-----  
Received: Temp=27°C | RSSI=-45 dBm | Success rate=80.00%  
-----  
Invalid value=15 | RSSI=-36 dBm | Success rate=66.67%  
-----  
Received: Temp=22°C | RSSI=-38 dBm | Success rate=71.43%  
-----  
Received: Temp=23°C | RSSI=-37 dBm | Success rate=75.00%  
-----  
Invalid value=29 | RSSI=-37 dBm | Success rate=66.67%  
-----  
Received: Temp=23°C | RSSI=-38 dBm | Success rate=70.00%  
-----  
Received: Temp=22°C | RSSI=-38 dBm | Success rate=72.73%
```

Figure 5.2 Receiver output with RIS

5.3.3. With Intelligent RIS

Observations:

- Highest RSSI values recorded
- Stable and consistent signal levels
- Significant improvement in packet success rate
- Reduced decoding errors

The intelligent RIS dynamically selects the best reflection pattern based on RSSI, enabling adaptive optimization. This leads to improved communication reliability and demonstrates the effectiveness of intelligent control in RIS-assisted systems.

```

Receiver Ready
Received: Temp=22°C | RSSI=-38 dBm | Success rate=100.00%
-----
Received: Temp=27°C | RSSI=-37 dBm | Success rate=100.00%
-----
Received: Temp=23°C | RSSI=-45 dBm | Success rate=100.00%
-----
Received: Temp=27°C | RSSI=-44 dBm | Success rate=100.00%
-----
Received: Temp=22°C | RSSI=-40 dBm | Success rate=100.00%
-----
Received: Temp=23°C | RSSI=-41 dBm | Success rate=100.00%
-----
Received: Temp=27°C | RSSI=-36 dBm | Success rate=100.00%
-----
Invalid value=31 | RSSI=-42 dBm | Success rate=87.50%
-----
Received: Temp=27°C | RSSI=-40 dBm | Success rate=88.89%
-----
Invalid value=31 | RSSI=-40 dBm | Success rate=80.00%
-----
Received: Temp=22°C | RSSI=-39 dBm | Success rate=81.82%
-----

```

Figure 5.3 Receiver output with intelligent RIS

The overall system performance across the three configurations is comparatively analyzed and summarized. Table 5.1 shows the performance comparison of each of the RIS configurations.

Table 5.1 Performance comparison of the RIS configurations

Configuration	RSSI Strength	Stability	Success Rate	Performance
Without RIS	Low (Average RSSI Strength = -47 dBm)	Medium	Low (Average success rate = 48.19%)	Weak
With RIS(Static)	Moderate (Average RSSI Strength = -43 dBm)	Medium	Moderate (Average success rate = 77.25%)	Improved
With Intelligent RIS	High (Average RSSI Strength = -39 dBm)	High	High (Average success rate = 95.64%)	Best

5.4. Key Observations

- 1) Battery-free sensing is achievable using energy harvesting (solar-based implementation).
- 2) Backscatter communication performance is strongly dependent on signal strength.
- 3) RIS significantly improves indoor communication conditions.
- 4) Intelligent RIS enables adaptive optimization and achieves the best performance.
- 5) System latency remains a limitation due to transmission timing design.

The results show that:

- RIS enhances signal propagation.
- Intelligent RIS provides adaptive optimization.
- Improved RSSI directly leads to better decoding accuracy and reliability.

5.5. Simulation results

This section presents the MATLAB simulation results evaluating the performance of the proposed RIS-assisted backscatter communication system in an indoor 6G-like environment. This compares three scenarios without RIS, with non-intelligent RIS, and with intelligent RIS, focusing on received signal strength (RSSI), signal-to-noise ratio (SNR), and link reliability.

The simulation was implemented in MATLAB, modeling a fixed indoor environment where the distances between the transmitter, sensor node, RIS, and receiver remain constant. Environmental variations such as obstacles and multipath effects were represented using random channel variations.

The following three cases were considered:

- Without RIS: Direct backscatter communication between the sensor and receiver.

- With RIS (Non-intelligent): RIS provides reflection but without optimization.
- With Intelligent RIS: RIS adaptively selects the optimal configuration to maximize received signal strength.

Signal propagation was modeled using a simplified indoor path loss model with wall attenuation and the received signal strength was calculated and used to determine SNR as:

$$SNR = P_r - P_n \quad (5.1)$$

where (P_r) is the received power and (P_n) is the noise floor.

Link reliability was defined as the fraction of trials where the SNR exceeds a predefined threshold:

$$Reliability = \frac{\text{Number of successful trials}}{\text{Total trials}} \quad (5.2)$$

5.5.1. RSSI Performance Analysis

The RSSI results show a clear improvement across the three scenarios as shown in

Figure 5.4. Without RIS, the received signal strength is significantly low due to uncontrolled propagation, path loss, and multipath fading. The signal experiences severe attenuation, resulting in weak and unstable reception.

With the introduction of a non-intelligent RIS, RSSI improves as the reflecting surface redirects part of the incident and backscattered signals toward the receiver. However, because the reflection pattern is fixed, the improvement is limited and does not always align with optimal signal paths.

In the intelligent RIS case, a significant increase in RSSI is observed. The system evaluates multiple reflection configurations and selects the one that maximizes constructive signal combination at the receiver. This adaptive behaviour reduces destructive interference and improves overall signal strength.

Overall, the average RSSI improves from approximately -68 dBm (without RIS) to -62 dBm (non-intelligent RIS) and further to -52 dBm (intelligent RIS).

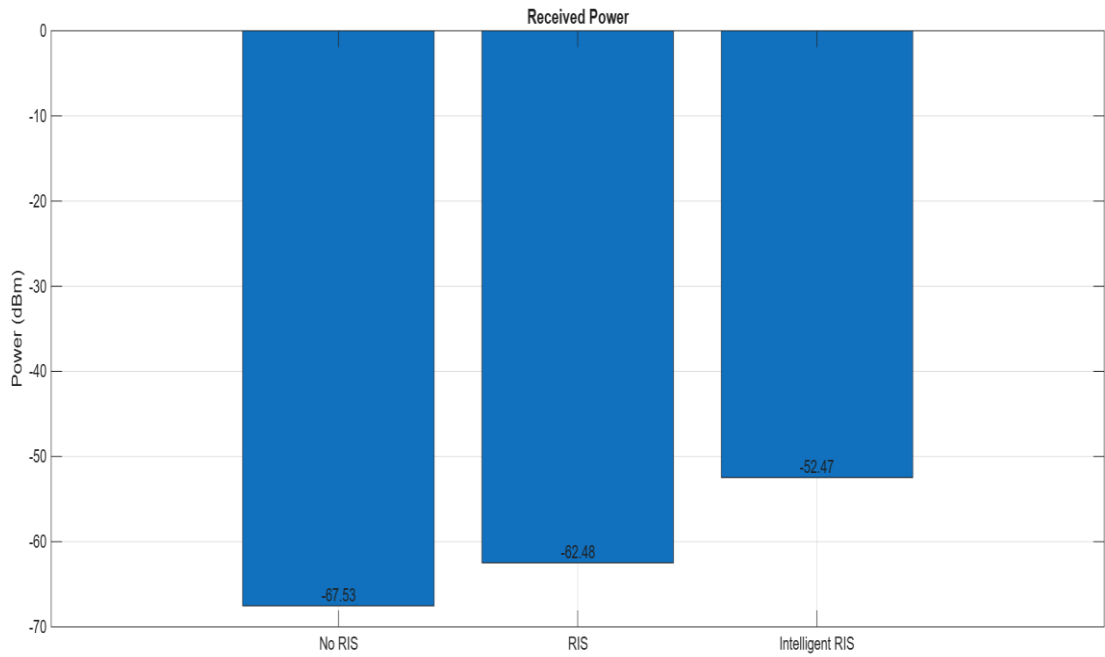


Figure 5.4 Average RSSI for each environment state

5.5.2. SNR Performance Analysis

The SNR results follow the same trend as RSSI as shown in Figure 5.5, since received power directly influences signal quality. In the absence of RIS, low received power results in poor SNR values, making the signal highly susceptible to noise and detection errors. This leads to unreliable communication performance.

With non-intelligent RIS, SNR improves due to increased reflected signal power, enhancing detectability at the receiver. However, the fixed configuration limits optimal signal enhancement.

With intelligent RIS, SNR reaches its highest values due to adaptive optimization of reflection patterns. By selecting the configuration that maximizes received power, the system significantly improves signal quality relative to the noise floor.

The average SNR improves from approximately 22 dB (without RIS) to 28 dB (non-intelligent RIS) and up to 38 dB (intelligent RIS).

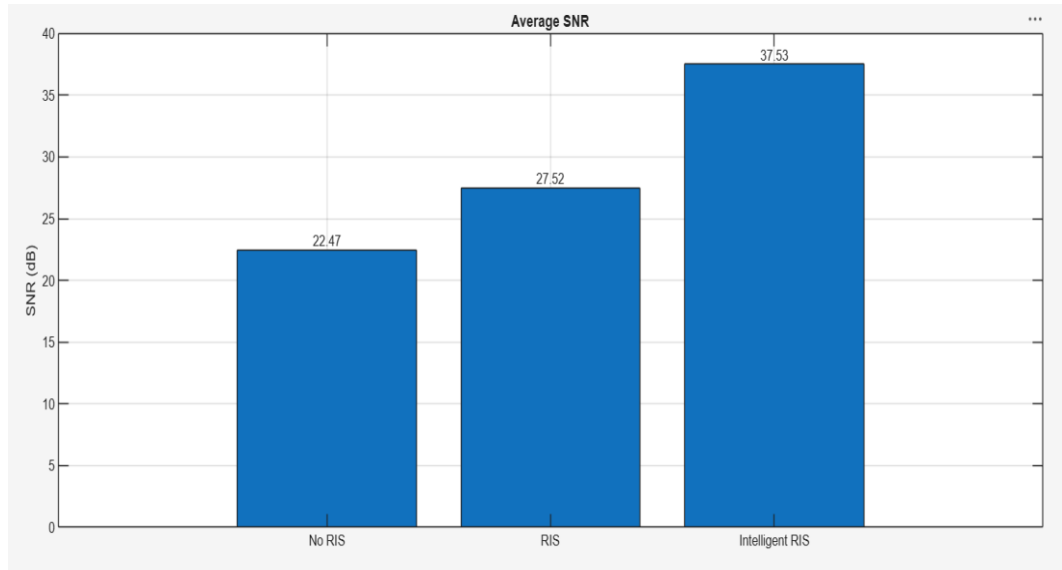


Figure 5.5 Average SNR for each environment state

5.5.3. Link Reliability Analysis

Link reliability reflects the percentage of successful communication trials where the received signal exceeds the detection threshold, this is shown in Figure 5.6. Without RIS, reliability is extremely low due to frequent SNR drops below the threshold, resulting in failed packet detection.

With non-intelligent RIS, reliability improves moderately as signal reinforcement increases the number of successful detections. However, performance remains inconsistent due to non-optimized reflection patterns.

With intelligent RIS, reliability reaches its maximum value, as adaptive configuration ensures that the received signal consistently exceeds the required threshold across trials.

The observed reliability improves from approximately 0.85 (without RIS) to 0.95 (non-intelligent RIS) and 0.98 (intelligent RIS).

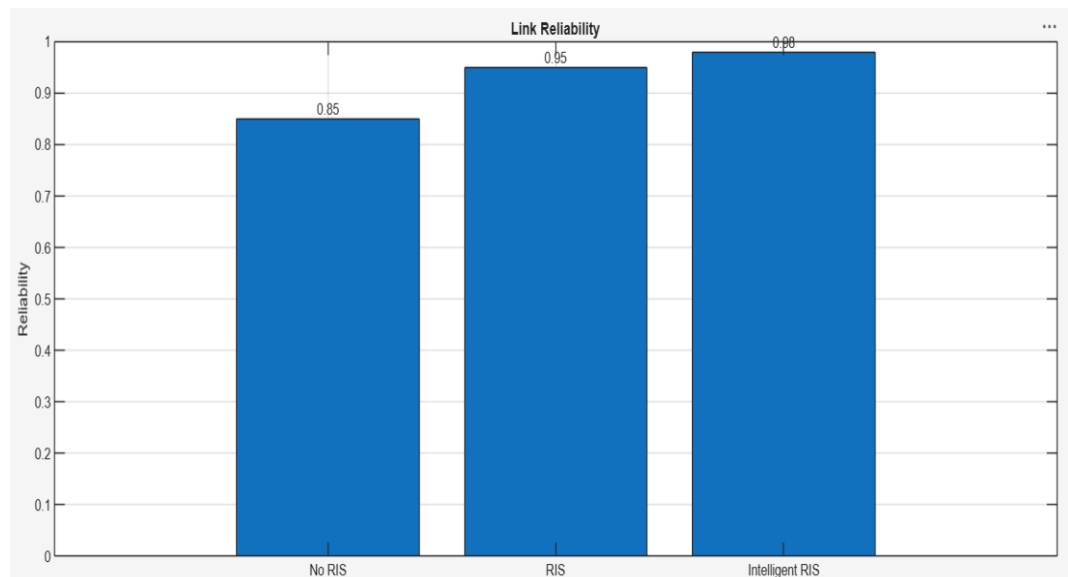


Figure 5.6 Link reliability for each environment state

5.5.4. Latency

===== LATENCY RESULTS =====		
Scenario	Latency_ms	Meets_Target
{ 'No RIS' }	36.14	true
{ 'RIS' }	34.107	true
{ 'Intelligent RIS' }	33.056	true

Figure 5.7 Latency results in the simulation

Figure 5.7 presents the latency performance of the proposed system under the No RIS, RIS, and Intelligent RIS scenarios. The Matlab simulation results show that all scenarios achieved latency values below the target threshold of 50ms, with intelligent RIS achieving the lowest latency. The reduction in latency is attributed to improved signal quality and reduced retransmissions in the RIS-assisted communication system.

The latency objective was not fully achieved in the physical prototype implementation, the difference between the simulation and prototype results is due to practical factors such as hardware processing delays, communication overhead, synchronization limitations, environmental interference, and power constraints associated with the battery-free sensor node. Although the prototype exceeded the target latency, the RIS-assisted configurations still showed improved performance relative to the non-RIS setup, thereby validating the effectiveness of the proposed approach under real-world operating conditions.

5.6. The Simulation Key Findings

- 1) Without RIS, the system suffers from severe signal degradation and low reliability.
- 2) Non-intelligent RIS improves signal strength but lacks consistency due to fixed configurations.
- 3) Intelligent RIS provides the best performance through adaptive optimization.
- 4) RSSI improvement directly leads to higher SNR and improved link reliability.
- 5) The results validate the effectiveness of RIS in enhancing backscatter communication systems in indoor environments.

The results demonstrate a clear relationship between RSSI, SNR, and link reliability. Improvements in RSSI directly enhance SNR, which in turn significantly increases communication reliability.

CHAPTER SIX

6. CONCLUSION AND RECOMMENDATIONS

This chapter presents the overall conclusion of the project based on the design, implementation, and evaluation of the intelligent RIS-assisted battery-free indoor sensor network. It also highlights the key challenges encountered during development and provides recommendations for future improvement.

6.1. Conclusion

The project successfully demonstrated the design and implementation of an intelligent Reconfigurable Intelligent Surface (RIS)-assisted indoor sensor network inspired by 6G communication concepts. The system integrates a battery-free sensor node, an RIS control unit, and an ESP32-based receiver to enable low-power backscatter communication. The results showed that the inclusion of RIS significantly improves signal propagation by enhancing reflected RF signals, leading to increased RSSI, improved SNR, and higher link reliability.

Furthermore, the introduction of an intelligent RIS control mechanism enabled adaptive selection of the optimal reflection pattern, resulting in the best overall communication performance compared to both non-RIS and static RIS configurations. Although the system successfully demonstrates reliable data transmission and environmental sensing, certain limitations such as latency constraints and RSSI variability were observed, which indicate areas for further optimization.

6.2. Challenges

Several challenges were encountered during the design and implementation of the system:

1) RF Energy Harvesting and Power Constraint

The implementation of RF energy harvesting faced the limitation that the harvested RF energy was insufficient to reliably power the MT3608 module, preventing the realization of a fully self-sustaining RF-powered system. To address the limitation, a solar-assisted power source was adopted to ensure stable and continuous operation of the sensor node. Despite this modification, the system still maintains its battery-free operational concept at the sensor level, as it does not rely on conventional battery storage for sensing and data transmission, but instead uses renewable energy harvesting as the primary power strategy.

2) Latency Constraint

Achieving the target end-to-end latency of ≤ 50 ms proved difficult. The system was limited by RSSI sampling delays in the ESP32 Wi-Fi stack, processing time for signal decoding and RIS configuration switching delays. As a result, the system operates at higher but still significantly reduced latency compared to the initial design.

3) RSSI Threshold Instability

The receiver node relies on RSSI thresholding to decode binary data. However, the threshold value was observed to fluctuate due to environmental variations, multipath effects, and Wi-Fi signal instability. This sometimes affected decoding accuracy and introduced occasional errors in reconstructed data.

4) Environmental Sensitivity

The system performance was found to be sensitive to indoor conditions such as distance between nodes, presence of obstacles and multipath fading effects. This affected consistency in RSSI measurements across different test runs.

6.3. Recommendations

To improve the performance and scalability of the system, the following recommendations are proposed:

1) Use of Advanced Signal Processing

To improve decoding accuracy replace RSSI-based detection with Channel State Information (CSI). CSI provides more detailed channel information and is less sensitive to environmental fluctuations, enabling more accurate detection of backscatter signals. Implement adaptive thresholding techniques instead of fixed thresholds and use digital filtering to reduce noise in signal readings.

2) Improved RIS Design

Increase number of controllable elements for finer control of signal reflections and implement phase-shift based RIS instead of binary amplitude control, this allows more precise manipulation of electromagnetic waves, leading to stronger signal enhancement, and better communication performance.

3) Receiver Robustness Improvement

Introduce error correction coding techniques such as hamming codes to detect and correct transmission errors caused by noise and signal fluctuations. In addition, advanced artificial intelligence optimization techniques such as machine learning or reinforcement learning can be implemented to enable more efficient and adaptive RIS configuration selection in dynamic indoor environments.

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APPENDIX

Appendix A: Project Budget

Item	Quantity	Unit Cost (Kshs)	Total Cost (Kshs)
Arduino Nano	1	2000	2000
ESP32	2	1500	3000
Temperature Sensor(DHT11)	1	300	300
Copper Wire	1	100	100
Aluminum Foil	1 roll	1000	1000
Solar Cell	1	1000	1000
MT3608 Module	1	150	150
Schottky Diodes (1N5819)	1	30	30
PIN Diodes	4	40	160
Transistors (2N2222)	1	50	50
Resistors	3	20	60
Capacitors	4	50	200
Jumper Wires	1 pack	200	200
Breadboards	2	500	1000
Total	9250		

Appendix B: Matlab Simulation Code

```

Command Window
>> clc; clear; close all;
%% =====
% 1. PARAMETERS
% =====
Pt = 0; % dBm
f = 2.4e9;
c = 3e8;
lambda = c / f;
noise_power = -90; % dBm
threshold = 10; % SNR threshold
num_trials = 1000;
% Storage
Pr_no_all = zeros(1,num_trials);
Pr_RIS_all = zeros(1,num_trials);
Pr_int_all = zeros(1,num_trials);

SNR_no_all = zeros(1,num_trials);
SNR_RIS_all = zeros(1,num_trials);
SNR_int_all = zeros(1,num_trials);

lat_no_all = zeros(1,num_trials);
lat_RIS_all = zeros(1,num_trials);
lat_int_all = zeros(1,num_trials);

success_no = 0;
success_RIS = 0;
success_int = 0;

for i = 1:num_trials

    d = 5 + (50-5)*rand; % random distance

    PL = 20*log10(4*pi*d/lambda);
    fading = randn * 2;

    % Received Power
    Pr_no = Pt - PL + fading;
    Pr_RIS = Pt - PL + (5 + randn) + fading;
    Pr_int = Pt - PL + (15 + 2*randn) + fading;
% Enforce ordering
if Pr_RIS <= Pr_no
    Pr_RIS = Pr_no + abs(randn);
end
if Pr_int <= Pr_RIS
    Pr_int = Pr_RIS + abs(randn);
end
% SNR
SNR_no = Pr_no - noise_power;
SNR_RIS = Pr_RIS - noise_power;
SNR_int = Pr_int - noise_power;
% Count success
success_no = success_no + (SNR_no > threshold);
success_RIS = success_RIS + (SNR_RIS > threshold);
success_int = success_int + (SNR_int > threshold);
%% ===== IMPROVED LATENCY MODEL =====
k = 0.4;
p_no = 1/(1 + exp(-k*(SNR_no - threshold)));
p_RIS = 1/(1 + exp(-k*(SNR_RIS - threshold)));
p_int = 1/(1 + exp(-k*(SNR_int - threshold)));
% Clamp probabilities
p_no = max(min(p_no,0.9),0.2);
p_RIS = max(min(p_RIS,0.95),0.3);
p_int = max(min(p_int,0.98),0.5);

prop_delay = d / c;
proc_delay = 0.03 + 0.005*rand;

lat_no = (prop_delay + proc_delay) / p_no;
lat_RIS = (prop_delay + proc_delay) / p_RIS;
lat_int = (prop_delay + proc_delay) / p_int;
% Store
Pr_no_all(i) = Pr_no;
Pr_RIS_all(i) = Pr_RIS;
Pr_int_all(i) = Pr_int;

SNR_no_all(i) = SNR_no;
SNR_RIS_all(i) = SNR_RIS;
SNR_int_all(i) = SNR_int;

lat_no_all(i) = lat_no;
lat_RIS_all(i) = lat_RIS;
lat_int = mean(lat_int_all)*1000;

%% =====
% 4. BAR GRAPHS WITH VALUES
% =====

labels = {'No RIS','RIS','Intelligent RIS'};

% --- Received Power ---
vals1 = [mean(Pr_no_all), mean(Pr_RIS_all), mean(Pr_int_all)];
figure;
b1 = bar(vals1);
set(gca,'XTickLabel',labels);
ylabel('Power (dBm)');
title('Received Power');
grid on;

for i=1:3
    text(i,vals1(i),sprintf('%.2f',vals1(i)),...
        'HorizontalAlignment','center','VerticalAlignment','bottom');
end

```

```

% --- SNR ---
vals2 = [mean(SNR_no_all), mean(SNR_RIS_all), mean(SNR_int_all)];
figure;
b2 = bar(vals2);
set(gca,'XTickLabel',labels);
ylabel('SNR (dB)');
title('Average SNR');
grid on;

for i=1:3
    text(i,vals2(i),sprintf('%.2f',vals2(i)),...
        'HorizontalAlignment','center','VerticalAlignment','bottom');
end

% --- Reliability ---
vals3 = [rel_no, rel_RIS, rel_int];
figure;
b3 = bar(vals3);
set(gca,'XTickLabel',labels);
ylabel('Reliability');
title('Link Reliability');
grid on;

for i=1:3
    text(i,vals3(i),sprintf('%.2f',vals3(i)),...
        'HorizontalAlignment','center','VerticalAlignment','bottom');
end

%% =====
% 5. LATENCY TABLE
% =====
Scenario = {'No RIS'; 'RIS'; 'Intelligent RIS'};
Latency_ms = [lat_no; lat_RIS; lat_int];
Meets_Target = Latency_ms <= 50;

Latency_Table = table(Scenario, Latency_ms, Meets_Target);

disp('===== LATENCY RESULTS =====');
disp(Latency_Table);

```